

ME23312 ENGINEERING THERMODYNAMICS

Unit II – Second Law and Availability Analysis · Presentation 3 of 10

The Carnot Cycle and the Reversed Carnot Cycle

Four reversible processes · p – V and T – s diagrams · Carnot efficiency · the reversed cycle and its COPs · why the Carnot cycle cannot be realised

In this presentation

1. Sadi Carnot's cycle
 2. The four processes
 3. The cycle on the p – V and T – s planes
 4. Efficiency of the Carnot cycle
 5. What the efficiency depends on
1. The reversed Carnot cycle
 2. COP of the reversed Carnot cycle
 3. Why the Carnot cycle cannot be realised in practice
 4. Why it still matters

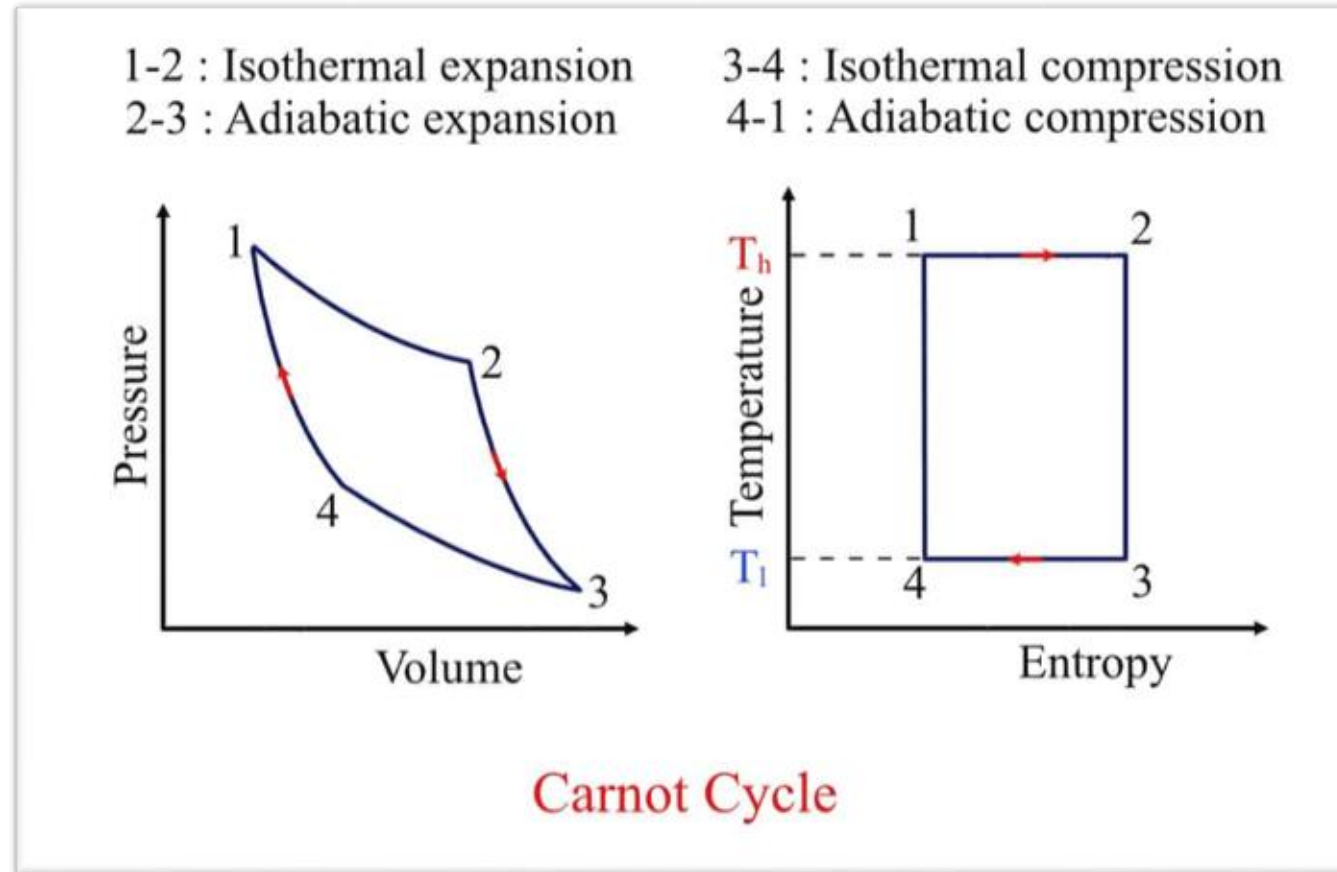
1

The Carnot cycle

Sadi Carnot's cycle (1824)

- The cycle with the **highest possible efficiency** for a heat engine between two reservoirs
- Four **reversible** processes executed by a working substance – here a gas in a piston–cylinder
- Two reversible isothermals + two reversible adiabatics

The Carnot cycle on the p–V plane



1–2 isothermal expansion at T_1 · 2–3 adiabatic expansion · 3–4 isothermal compression at T_2 ·
4–1 adiabatic compression

The four processes (1)

1–2 Reversible isothermal expansion at T_1

Cylinder in contact with the source; the gas expands slowly, absorbing heat Q_1 while its temperature stays at T_1 . For an ideal gas $Q_1 = W_{1-2} = mRT_1 \ln (V_2/V_1)$.

2–3 Reversible adiabatic expansion

Cylinder insulated; the gas continues to expand, doing work at the expense of its internal energy, until its temperature falls to T_2 .

The four processes (2)

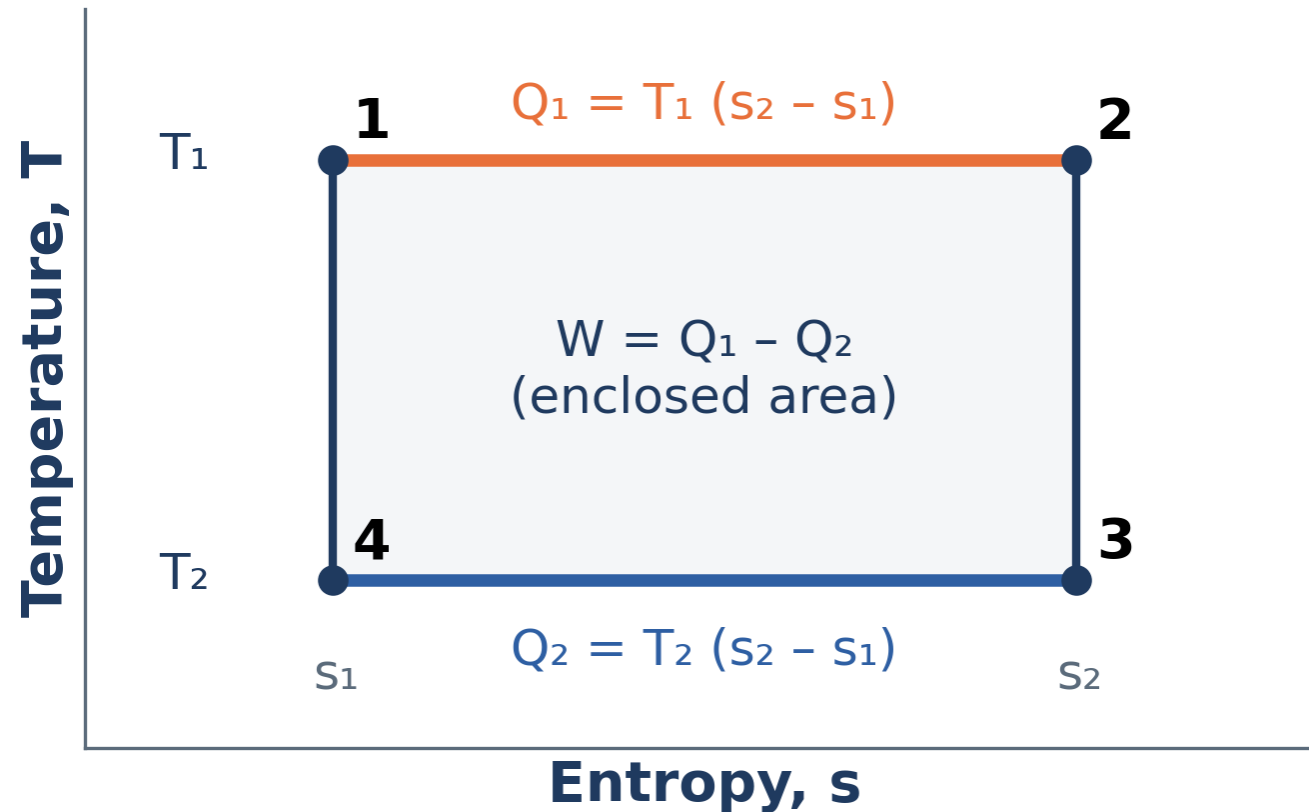
3–4 Reversible isothermal compression at T_2

Cylinder in contact with the sink; the gas is compressed slowly, rejecting heat $Q_2 = mRT_2 \ln (V_3/V_4)$ to the sink at constant T_2 .

4–1 Reversible adiabatic compression

Cylinder insulated again; the gas is compressed until its temperature rises to T_1 and it returns to state 1.

The Carnot cycle on the T-s plane



A rectangle: $Q_1 = T_1 (s_2 - s_1)$ is the area under 1–2, $Q_2 = T_2 (s_2 - s_1)$ the area under 3–4, $W = Q_1 - Q_2$ the enclosed area

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Efficiency of the Carnot cycle

The Crucial Step: Compare Q_h and Q_l

The equal entropy changes come from the two reversible adiabatic processes

$$Q_h = T_h(S_2 - S_1)$$

$$Q_l = T_l(S_3 - S_4)$$

$$S_2 = S_3 \quad \text{and} \quad S_1 = S_4$$

$$\text{Therefore: } S_2 - S_1 = S_3 - S_4 = \Delta S$$

$$Q_h = T_h \Delta S \quad \text{and} \quad Q_l = T_l \Delta S$$

$$Q_l / Q_h = (T_l \Delta S) / (T_h \Delta S) = T_l / T_h$$

Derive the Carnot Efficiency

Start from the definition of heat-engine efficiency

$$\eta = \text{Work output} / \text{Heat supplied}$$

$$\eta = W / Q_h$$

$$W = Q_h - Q_l$$

$$\eta = (Q_h - Q_l)/Q_h = 1 - Q_l/Q_h$$

$$\text{Since } Q_l/Q_h = T_l/T_h$$

$$\eta_{\text{Carnot}} = 1 - T_l/T_h$$

Carnot efficiency

$$\eta_{\text{Carnot}} = 1 - T_2/T_1 = (T_1 - T_2)/T_1 \quad (T \text{ in kelvin})$$

- From the T–s rectangle directly: $\eta = W/Q_1 = (T_1 - T_2)(s_2 - s_1) / [T_1 (s_2 - s_1)]$
- Depends **only on the two reservoir temperatures** – not on the working substance

What the efficiency tells us

- Increases as T_1 is raised or T_2 is lowered
- Equals unity only if $T_2 = 0$ K or $T_1 \rightarrow \infty$ – neither is attainable
- Equals zero if $T_1 = T_2$ – no work from a single reservoir, as Kelvin–Planck says

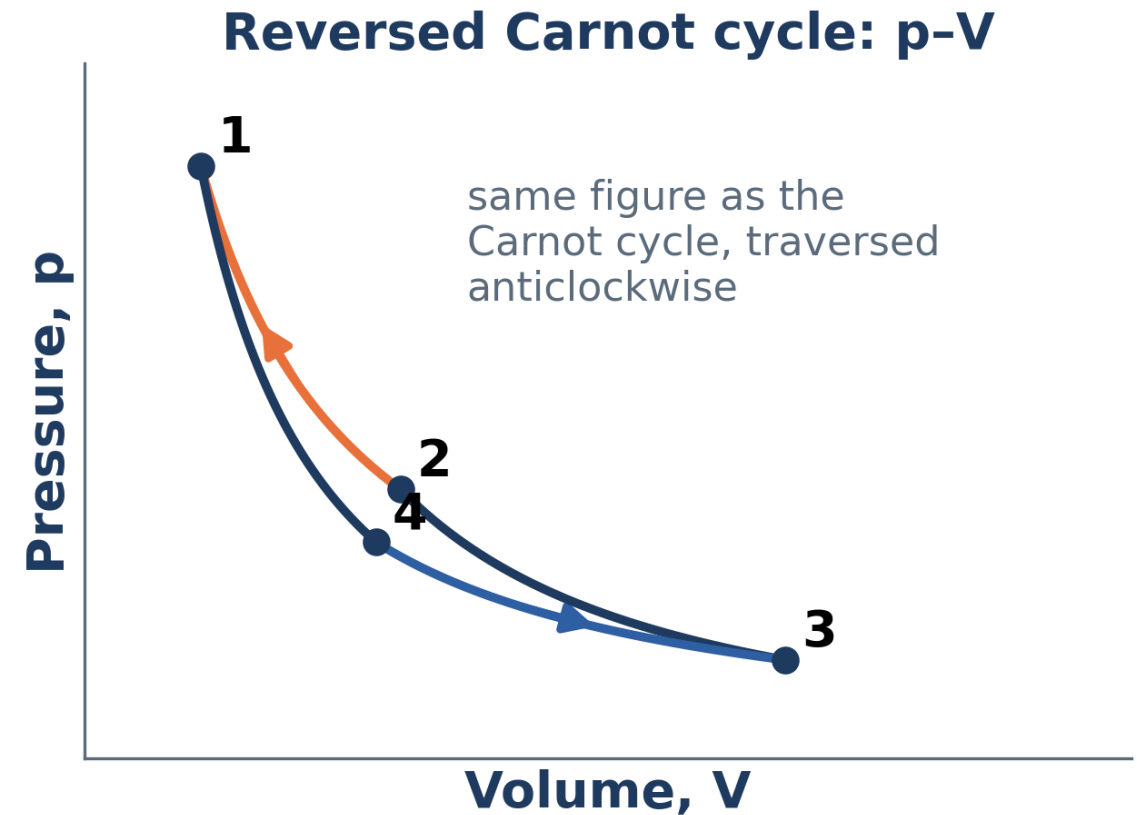
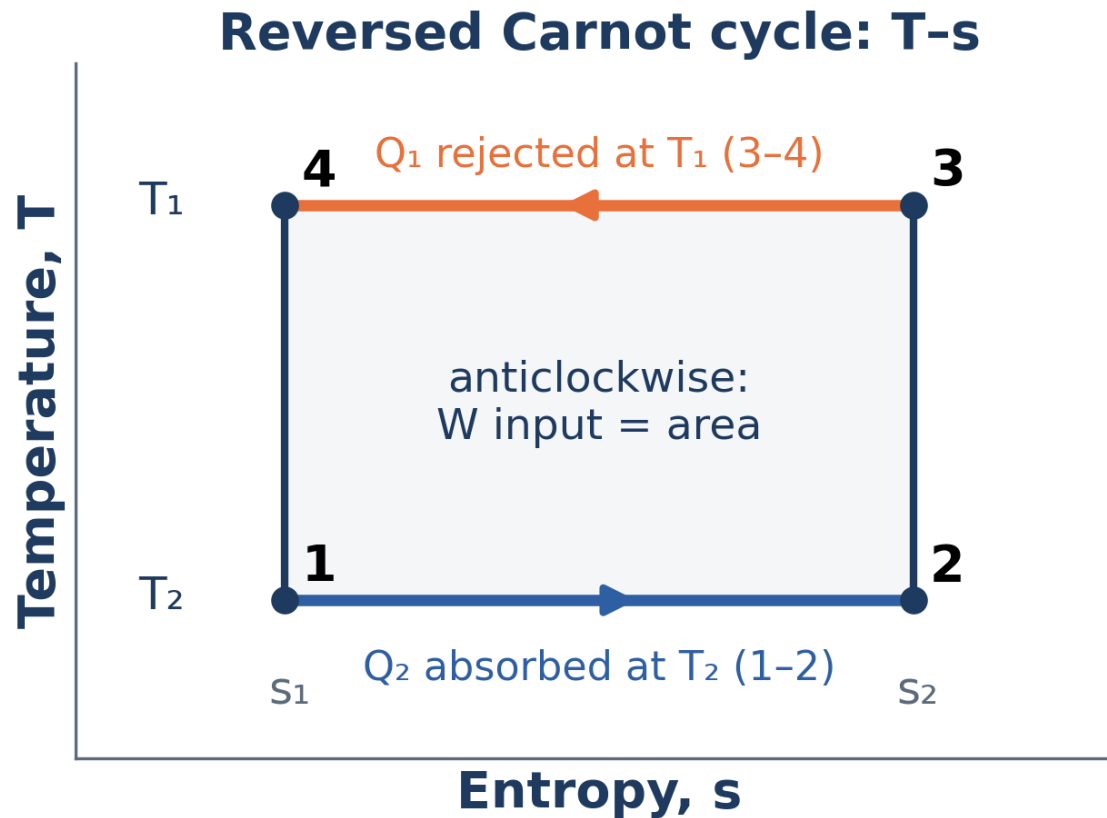
KEY POINT Always use kelvin: a ratio such as T_2/T_1 is meaningless in °C.

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The reversed Carnot cycle

The ideal refrigerator and heat pump

The reversed Carnot cycle



The same figure traversed anticlockwise 1-2-3-4-1; the enclosed area is now the work input

Processes of the reversed cycle

1. 1–2 isothermal expansion at T_2 : heat Q_2 absorbed from the cold body
2. 2–3 reversible adiabatic compression: temperature rises from T_2 to T_1
3. 3–4 isothermal compression at T_1 : heat Q_1 rejected to the hot body
4. 4–1 reversible adiabatic expansion: temperature falls back to T_2

KEY POINT Net work input = enclosed area = $Q_1 - Q_2 = (T_1 - T_2)(s_2 - s_1)$

COP of the reversed Carnot cycle

$$\begin{aligned} (\text{COP})_{\text{ref,Carnot}} &= Q_2 / (Q_1 - Q_2) = T_2 / (T_1 - T_2) \\ (\text{COP})_{\text{HP,Carnot}} &= Q_1 / (Q_1 - Q_2) = T_1 / (T_1 - T_2) \end{aligned}$$

- The highest COP possible between T_1 and T_2
- $(\text{COP})_{\text{HP}} - (\text{COP})_{\text{ref}} = 1$, as for any refrigerator–heat pump pair

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Why the Carnot cycle cannot be realised

Practical obstacles (1)

Infinitely slow and infinitely fast

Isothermal heat transfer needs an infinitesimal ΔT – an infinitely slow process (or infinite heat-transfer area); the adiabatics must be very fast to avoid heat loss. The same piston cannot be both.

Tiny work per cycle

Isothermal and adiabatic curves on the p – V plane have nearly the same slope (1 and $\gamma = 1.4$ for air), so the enclosed area is very small – an engine of practical size develops little power.

Practical obstacles (2)

Ideal hardware

A frictionless piston; a cylinder that is a perfect insulator in two strokes and a perfect conductor in the other two; a working substance heated and cooled without loss.

Two-phase compression

In the reversed cycle with a refrigerant the compressor cannot handle a liquid–vapour mixture, and expanding the liquid in a turbine recovers very little work.

Why the Carnot cycle still matters

It sets the upper limit of performance of any engine, refrigerator or heat pump working between two temperatures.

- Every practical cycle – Rankine, Otto, Diesel, Brayton, vapour compression – is judged by how closely it approaches the Carnot value

Summary

- Carnot cycle: reversible
isothermal at T_1 , adiabatic,
isothermal at T_2 , adiabatic
- T–s plane: a rectangle; heats are
areas under the isotherms, work
the enclosed area
- $\eta_{\text{Carnot}} = 1 - T_2/T_1$ – depends on
temperatures only
- Reversed cycle: $(\text{COP})_{\text{ref}} = T_2/(T_1 - T_2)$, $(\text{COP})_{\text{HP}} = T_1/(T_1 - T_2)$
- Not practicable: infinitely slow
isotherms, tiny work per cycle,
ideal hardware
- The benchmark for every real
cycle

Next: The Carnot theorem and the Kelvin scale

- Statement and proof of the Carnot theorem; its corollaries
- The thermodynamic temperature scale, $(Q_1/Q_2)_{\text{rev}} = T_1/T_2$
- Numerical problems: Worked Examples 2.1 – 2.6 in the lecture notes